Rebecca L. Martinez

Undergraduate Researcher | Student Assistant
Environmental Studies Program | University of California, Santa Barbara
rebecca_martinez@ucsb.edu | martinez.rl@hotmail.com

Education

B.S. in Environmental Studies, University of California, Santa Barbara — Expected 2027
A.S. in Biology, Allan Hancock College *(With honors)* — Jun 2024
A.A. in Biology, Allan Hancock College *(With honors)* — Dec 2023
A.A. in Mathematics and Science, Allan Hancock College *(With high honors)* — Jun 2021

Awards and Honors

Fall Birnam Wood Veterans Scholarship — $3,000 — 2026
Fall Birnam Wood Veterans Scholarship — $3,000 — 2025
Gene & Susan Lucas Undergraduate Research Fund — $500 — 2025
Crankstart Transfer Scholarship — $5,000 — 2025
2025 UCSB Edison STEM Conference Research Scholarship — $2000 — 2025
UCNRS Field Science Fellowship — $5,000 + $1,000 project stipend — 2025
Bren Environmental Studies Fellows Program — $600 — 2025
Gene & Susan Lucas Undergraduate Research Fund — $1,000 — 2025
Wenger Scholarship, UCSB Environmental Studies — $2,000 — 2025
UCSB Promise Scholarship — $60,000 total over 2 years — 2024
PEO Foundation Scholarship — $1,250 — 2024
UCSB MESA M-STP Program Scholarship — $500 — 2024
Capt. G. Allan Hancock Scholarship — $3,500 — 2024
Allan Hancock College MESA Scholarship — $500 — 2023
Brander Vineyard Scholarship — $3,000 — 2023
Brander Vineyard Scholarship — $1,000 — 2022
Joan Semelsberger-Roth Scholarship — $2,000 — 2022
Joan Semelsberger-Roth Scholarship — $1,000 — 2021
Alpha Gamma Sigma Kathleen D. Loly Academic Award — $800 — 2021
Alpha Gamma Sigma Aquarius Chapter Leadership Award — 2021
Joyce Dendo Veteran Scholarship — $500 — 2021
Virginia B. Martinez Memorial Scholarship — $500 — 2021

Research Experience

Undergraduate Researcher, SWAMP Lab, UC Santa Barbara — Jan 2025-Present
Geologic Nitrogen Inputs to Semi-Arid Soils with and without Mammalian Bioturbation

Conduct research with Dr. Iris Holzer on rock-derived nitrogen and mammalian bioturbation in semi-arid soils at Sedgwick Reserve and Santa Cruz Island. Collected and prepared soil and rock samples for isotope analysis using a rock saw and shatterbox, maintained project records, and analyzed data using R/RStudio.Presented project findings at the 2025 American Geophysical Union Annual Meeting.

Research Assistant, UC Santa Barbara — Mar 2025-Present
Soil Carbon Stabilization Study

Assisted with soil-pit excavation, bulk-density sampling, and soil-auger collection at UCNRS Sedgwick Reserve.

Completed USDA soil profile descriptions and followed field-sampling protocols.

Participant, UCSB–Smithsonian Scholars Program — Jun 2023-Aug 2023
Completed advanced field methods and data science training at UCNRS Santa Cruz Island Reserve and Smithsonian Tropical Research Institute (STRI), Panama, including RStudio/PositCloud instruction.

Professional Experience

Seasonal Park Aide, California State Parks, Channel Islands District — Sep 2026-Present
Staff the entrance station, collect day-use fees, register campers, issue passes, and answer visitor questions at Refugio and El Capitan State Beaches.

Operate cash-register and computer systems, explain park rules, and assist with general cleaning and park operations.

Student Assistant, UCSB Office of Education Partnerships — Oct 2024–June 2026
Collect data, maintain field sites, repair equipment, updated protocols, and supported project logistics to improve research operations.

Chumash Casino Resort, Santa Ynez, CA — Jun 2006–May 2025

Slot Attendant; assisted guests and resolved service and slot-equipment issues in a busy casino environment. — 2012-May 2025

Floor Operations Supervisor; supervised daily floor operations, coordinated staffing, and addressed guest and equipment issues while following casino procedures. — 2007-2012

Security Officer; responded to safety concerns and supported a safe environment for guests and employees. — Jun 2006–2007

Bulk Fuel Specialist (MOS 1391), United States Marine Corps Reserve — Jul 2005-Aug 2010

Teaching and Mentorship

Teaching Assistant, UCSB MESA Successful Transition Program (M-STP) — Sep 2025-Sep 2025
Supported a 3-week virtual chemistry workshop for incoming and transfer STEM students. Assisted with Zoom lectures, evaluated assignments, and mentored students to support their transition to UCSB.

Discussion Leader, UCSB–Smithsonian ERES Program — Jun-Sep 2025
Assisted udergraduate instruction of RStudio/PositCloud for data manipulation and visualization, mentored students during field workshops at Santa Cruz Island Reserve; trained students on camera traps, Audiomoths, and GPS.

Data Science Intern, Smithsonian Data Science Lab — Jun-Sep 2024
Taught R/RStudio and developed instructional resources to help undergraduate students work with data and research projects.

Teaching Assistant, Allan Hancock College Human Anatomy Open Lab — Aug 2023–Jun 2024
Facilitated Friday lab sessions, including setup and breakdown; developed visual study guides, led office hours, and supported human cadaver lab activities.

Tutor, Academic Resource Center & STEM Center, Allan Hancock College — Aug 2019–Jun 2024
One-on-one, drop-in, and group sessions. Subjects tutored: anatomy, algebra, botany, introductory biology, calculus 1 & 2, cellular biology, introductory & general chemistry, microbiology, and physiology.

Lead Tutor; trained new tutors, led workshops on study skills and time management, and supported onboarding of staff and faculty. — Aug 2020– Jun 2024

Embedded Tutor; Hosted review sessions and office hours, helped with grading, and developed study materials for life and physical science courses — Jan 2020– Jun 2024

Leadership and Service

Student Representative, UC Systemwide Surveys Student Advisory Group — Sep 2025–Present
Gained insight into the full survey process, collaborated on strategies to increase student participation,and provided feedback on survey initiatives

Outreach Volunteer, UCSB Veteran Support Center — Jan 2024–Present
Organized networking events to connect veteran students and women veterans within UCSB's communities.

Science Outreach Volunteer, Allan Hancock College Life & Physical Sciences Department — Jan 2022–Jun 2024
Helped design and facilitate hands-on science activities for local K–12 students; created educational materials and coordinated event logistics.

Student Representative, Allan Hancock College — Aug 2021–Jun 2023
Served as a student representative for faculty and staff hiring committees and Title V Committee to advocate for student needs.

Permanent Member, Alpha Gamma Sigma Aquarius Honor Society, Allan Hancock College — Aug 2019–Present

Vice President; Organized meetings, managed member eligibility and service records, and developed an online system to track service points. — Jan 2020–Jun 2021

Community Service Chair; Led service projects and fundraisers, organized initiatives including coat and sock drives, book and toy drives, and community clean-up events — Aug 2022–Jun 2023

Key Volunteer, Santa Barbara County Toys for Tots — Nov 2018–Present
Coordinated donation efforts and distributed toys to children during holiday seasons.

United States Marine Corps — Jun 2005–Oct 2010
Developed strong discipline, adaptability, and problem-solving under high-pressure conditions.

Presentations

Geologic Nitrogen Inputs to Semi-Arid Soils with and without Mammalian Bioturbation, American Geophysical Union (AGU) 2025 New Orleans, LA. — Dec 2025

Learning from the Ground Up: Mammalian Bioturbators, Soil, & Undergraduate Education, UCSB Environmental Studies 55th Anniversary, Santa Barbara, CA. — Apr 2025

Guest Speaker, Allan Hancock Foundation Honors Gala, Santa Maria, CA — May 2024

Comparison of Species Richness & Functional Richness Between Barro Colorado Island and Pipeline Road in Panama, Cal Poly Summer Internship Research Symposium, San Luis Obispo, CA. — Aug 2023

Comparison of Species Richness & Functional Richness Between Barro Colorado Island and Pipeline Road in Panama, UCSB Fall Undergraduate Research Showcase, Santa Barbara, CA. — Nov 2023

Programs and Affiliations

Bren–Environmental Studies Fellows Program — 2025–2026
Tau Sigma National Honor Society — 2025–Present
American Geophysical Union (AGU) — 2025–Present
UCSB Promise Scholars — 2024–Present
UCSB Letters & Science Honors Program — 2024–Present
UCSB Student Veterans Association (SVA) — 2024–Present
PEO Foundation — 2024–2025
UCSB–Smithsonian Scholars Program — 2023–2024
Association of Colleges for Tutoring & Learning Assistance (ACTLA) — 2020–Present

Alpha Gamma Sigma Honor Society	2019–Present
Lompoc Veteran Advisory Committee (LVAC)	2018–Present

Last updated: October 25, 2026

`https://rebeccalmartinez.github.io/personal_website/`